# AI POWERED FLEET PERFORMANCE AND DELIVERY EFFICIENCY DASHBOARD

## PROJECT REPORT:

**AI Powered Fleet Performance and Delivery Efficiency Dashboaard**

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## PROJECT OBJECTIVE

To evaluate fleet performance and delivery efficiency using Power BI  focusing on **on-time delivery**, **fuel performance**, and **operational cost optimization**.

## PROJECT OVERVIEW

The **Fleet Performance & Delivery Efficiency Dashboard** provides a comprehensive analysis of logistics operations using Power BI. It integrates trip and vehicle data to measure **on-time delivery performance**, **fuel efficiency**, and **operational costs** across routes and vehicle types. Through interactive visuals, KPIs, and AI-powered insights, the dashboard helps identify factors influencing delivery delays, monitor maintenance expenses, and optimize fleet utilization. This data-driven approach enables logistics managers to make informed decisions, reduce costs, and improve overall delivery efficiency.

## DATA MODELING

After completing data preprocessing, a robust **data model** was developed in Power BI to enable dynamic analysis across trips, vehicles, and delivery metrics. The model integrates both tables through their common key field and supports multiple calculated measures for performance tracking.

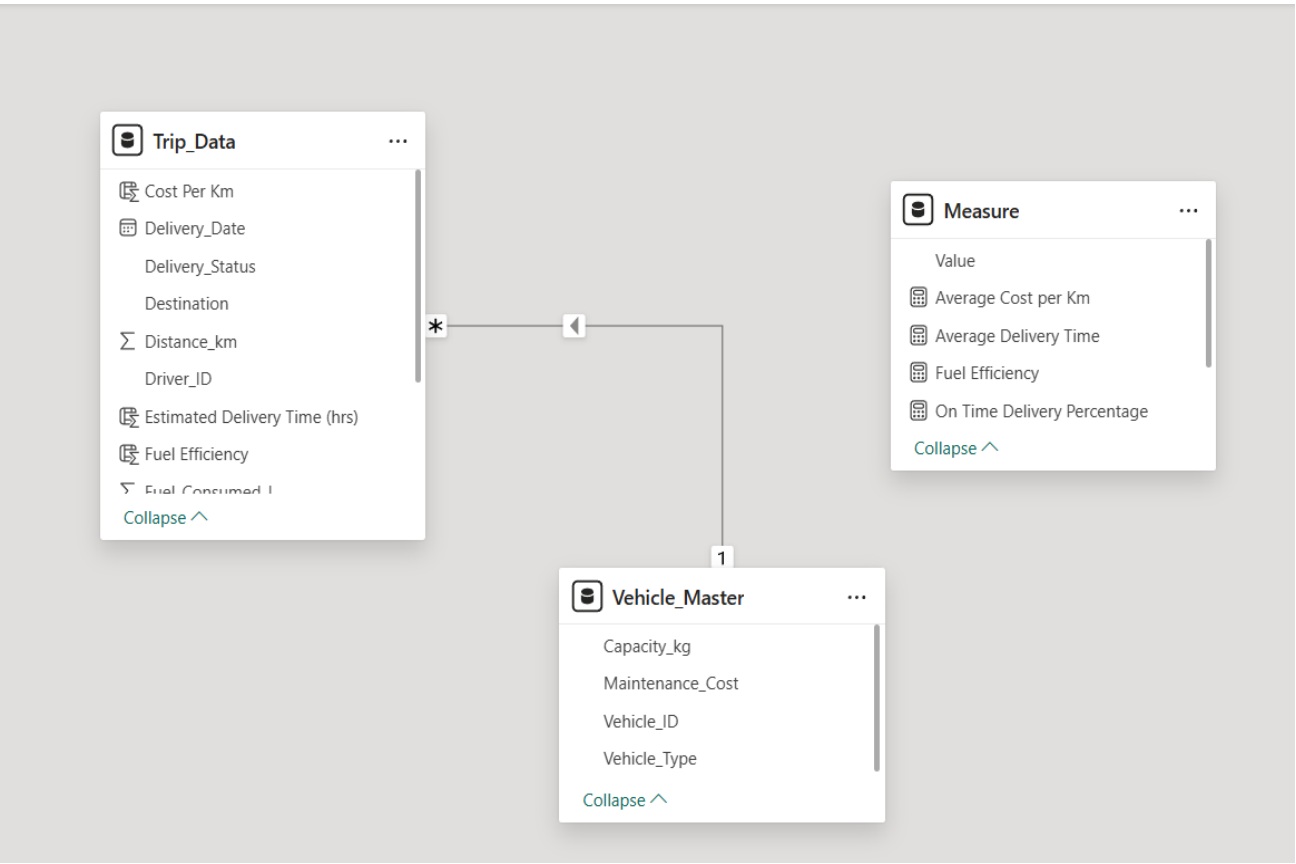
**1. Data Modeling**

- **Tables Used:**
    - Trip_Data: Contains trip details such as distance, fuel consumed, delivery date, and status.
    - Vehicle_Master: Contains vehicle information such as vehicle type, capacity, and maintenance cost.

- **Relationships:**
    - A **one-to-many relationship** was created:
    - Vehicle_Master[Vehicle_ID] → Trip_Data[Vehicle_ID]

This ensures that each vehicle's maintenance cost and attributes can be linked to multiple trips.

- **Model Validation:**
    - Ensured referential integrity (each Vehicle_ID in Trip_Data exists in Vehicle_Master).
    - Verified relationship direction to support correct filtering and aggregation.




**2. DAX Formulas**

Several **calculated columns and measures** were created using DAX (Data Analysis Expressions) to analyze fleet performance metrics.

## DATA VISUALIZATION

The Power BI dashboard visually represents key performance indicators (KPIs) and trends in fleet operations, providing actionable insights for logistics optimization. A mix of traditional

charts, KPI cards, and AI-powered visuals were used to analyze delivery efficiency, fuel usage, and cost performance.

**1. Visualizations Used**

| Visualization Type | Fields Used | Purpose |
|---|---|---|
| **Bar Chart** | Destination, On-Time Delivery % | Compare delivery punctuality across destinations |
| **Line Chart** | Delivery_Date, Fuel Efficiency | Show changes in fuel efficiency over time |
| **Cards** | Average Delivery Time (hrs), Average Cost per km | Display key performance metrics |
| **Pie Chart** | Vehicle_Type, Average(Maintenance_Cost) | Compare maintenance cost distribution among vehicle types |

**2. Key Dashboard Components**

**KPI Cards**

- **Average Delivery Time (hrs):** Displays the average estimated delivery duration across all trips.
- **Average Cost per km:** Represents the average operational cost per kilometer, combining fuel and maintenance expenses.

**Fuel Efficiency Trend (Line Chart)**

- Tracks how fuel efficiency varies by date, helping identify days or routes with low mileage.
- Assists in optimizing vehicle usage and route planning.

**On-Time Delivery Performance (Bar Chart)**

- Compares destinations based on on-time delivery percentage.
- Highlights high-performing routes and those needing improvement.

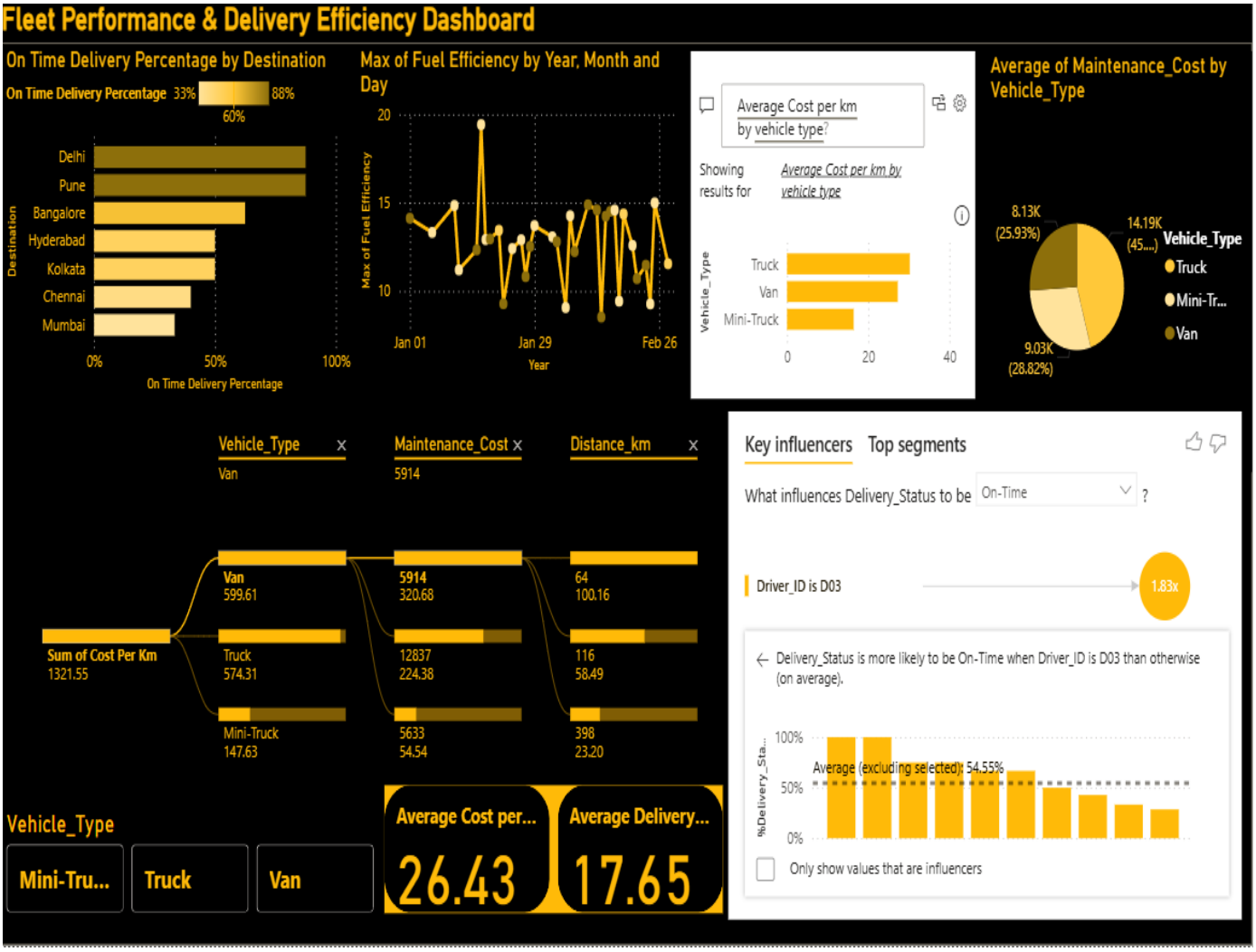
**Maintenance Cost Analysis (Pie Chart)**

- Visualizes the proportion of maintenance expenses across vehicle types.
- Helps identify which vehicle categories incur the highest upkeep cost.

### 3. AI-Powered Visuals

| AI Visual | Configuration | Insight |
|---|---|---|
| **Q&A Visual** | Prompt: "Average Cost per km by vehicle type?" | Allows users to ask natural-language questions and get instant insights. |
| **Key Influencers** | Analyze: Delivery_Status; Explain by: Distance_km, Vehicle_Type, Driver_ID | Identifies which variables most influence on-time or late deliveries. |
| **Decomposition Tree** | Analyze: Cost per km; Explain by: Vehicle_Type, Maintenance_Cost, Distance_km | Breaks down the factors contributing to higher or lower trip costs. |

The **Fleet Performance & Delivery Efficiency Dashboard** provides valuable insights into operational trends, delivery reliability, and vehicle utilization. These insights help the logistics company make informed decisions to improve performance and reduce costs.



# DASHBOARD INSIGHTS

**1. Delivery Performance Insights**

- **On-Time Deliveries:**
  A majority of trips were completed on time, with destinations such as **Pune** and **Delhi** achieving the highest punctuality rates.
- **Delayed Routes:**
  Routes like **Mumbai** and **Chennai** show a slightly higher delay rate, likely due to longer distances or traffic congestion.
- **Driver Impact:**
  Certain drivers consistently achieve better on-time delivery performance, indicating potential best practices or efficient driving patterns.

**2. Fuel Efficiency Insights**

- **Efficiency Range:**
  Fuel efficiency varies between **8–20 km/L**, showing that route type and vehicle condition play a major role in fuel consumption.
- **Performance Pattern:**
  **Mini-Trucks** and **Vans** demonstrate higher fuel efficiency compared to large trucks.
- **Optimization Potential:**
  Regular vehicle maintenance and route scheduling can help stabilize efficiency across all trips.

**3. Cost Insights**

- **Cost per km:**
  The **average cost per kilometer** is approximately **₹25–30**, depending on vehicle type and maintenance levels.
- **Maintenance Cost Influence:**
  Vehicles with higher maintenance costs (like trucks) significantly impact total trip cost.
- **Efficiency-Cost Relationship:**
  Routes with lower fuel efficiency tend to have higher cost per km, emphasizing the need for route planning and vehicle health monitoring.

**4. Vehicle & Maintenance Insights**

- **Maintenance Distribution:**
  Pie chart analysis shows **trucks** incur the highest maintenance cost share (~40–45%), while **mini-trucks** and **vans** are more economical.

- **Vehicle Utilization:**
  Frequent short-distance trips improve cost efficiency and on-time rates for smaller vehicles.

**5. AI-Driven Insights**

- **Key Influencers:**
  Delivery success is strongly influenced by **Vehicle_Type**, **Driver_ID**, and **Distance_km**.
- **Decomposition Tree:**
  Reveals that higher **maintenance cost** and longer **distance** are the top contributors to increased cost per km.
- **Q&A Visual:**
  Allows users to dynamically explore metrics like *"Average cost per km by vehicle type"* for deeper insights.

**Overall Insight Summary**

- **Operational Efficiency:** Medium trips with moderate distances yield the best cost-to-efficiency ratio.
- **Performance Drivers:** Driver behavior, vehicle type, and route length are the main influencers of delivery outcomes.
- **Optimization Focus:** Improving fuel efficiency and managing maintenance costs can

# CONCLUSION

The **AI-Driven Fleet Performance & Delivery Efficiency Dashboard** successfully integrates trip and vehicle data to deliver comprehensive insights into logistics operations.
Through **data modeling**, **DAX measures**, and **interactive Power BI visuals**, the dashboard highlights critical aspects such as on-time delivery performance, fuel efficiency, maintenance costs, and overall operational expenditure.

The use of **AI-powered visuals** (Q&A, Key Influencers, and Decomposition Tree) enhances the analytical depth, enabling the identification of hidden patterns and influencing factors behind fleet efficiency.
Overall, this Power BI solution empowers logistics managers to make **data-driven decisions**, optimize fleet utilization, and improve customer satisfaction through timely and cost-effective deliveries.

# RECOMMENDATIONS

1. **Enhance Data Tracking**
    - Include additional fields such as **Dispatch Date**, **Delivery Time**, and **Route Type** to calculate real-time delivery durations and improve delay analysis.
2. **Fuel Optimization**
    - Introduce GPS-based tracking and telematics to monitor driving behavior and identify routes with poor fuel efficiency.
3. **Maintenance Scheduling**
    - Implement predictive maintenance models using historical cost and distance data to reduce breakdowns and unplanned expenses.
4. **Driver Performance Monitoring**
    - Use driver-level KPIs (on-time %, average efficiency) to recognize top performers and provide targeted training for underperformers.
5. **Route Planning Automation**
    - Integrate AI-based route optimization tools with Power BI to automatically suggest the most efficient paths considering distance, traffic, and cost.
6. **Regular Dashboard Updates**
    - Refresh the dashboard weekly to maintain data accuracy and provide real-time insights for operational decision-making.

**Final Outcome**

The project demonstrates how **Power BI can transform raw logistics data into actionable intelligence**, supporting strategic decisions in fleet management.
By combining data analytics with AI-driven insights, the dashboard enhances visibility, reduces costs, and promotes overall delivery efficiency — creating a foundation for **smart, sustainable logistics operations**.